The instrument makes the winner: reconciling two contradictory 2026 head-to-head evaluations of clinical AI

Author: Koyar Afrasyab, M.D. — *corresponding author*. **Affiliation:** Independent researcher; Founder, Kinvectum AB. **Funding:** Kinvectum AB. **Correspondence:** Koyar Afrasyab (Kinvectum AB). ORCID: [to complete before submission]. **Article type:** Brief report / methodological reconciliation. **Preprint + code:** see *Data and code availability*.

Abstract

Background. Two 2026 head-to-head studies reached opposite conclusions about specialized clinical AI. Real-POCQi (arXiv:2606.28960) found OpenEvidence (OE) beat frontier general-purpose LLMs on blinded physician **pairwise preference** across 620 real point-of-care queries; a Nature Medicine study (s41591-026-04431-5) found frontier LLMs beat OE and UpToDate Expert AI under **absolute 1–4 rubric** scoring. Each study also sourced its "real-world" queries from the platform that won (home-field provenance), and the two used different model versions and evaluation instruments, confounding any direct comparison.

Objective. Test whether the **evaluation instrument alone** — with queries and answers held fixed — can flip the winner, isolating it from query provenance, model version, and answer content.

Methods. We reused Real-POCQi's public data (CC BY 4.0): the same 620 queries, the same four systems' verbatim answers (Claude Opus 4.8, Gemini 3.1 Pro, GPT-5.5, OpenEvidence), and the same blinded human pairwise ratings. On a seed-fixed 150-query sample we re-scored every answer with a **Nature-style absolute 1–4 rubric** administered by a blinded four-model LLM-judge panel (GPT-5.5, Claude Opus 4.8, Grok-4.3, Gemini-3.5-flash), all at **high reasoning effort with token consumption verified**. We derived the identical OE-vs-rest win-difference metric under both instruments and compared them, with 2,000-replicate cluster bootstrap (on `question_id`), per-judge self-preference, leave-one-judge-out robustness, and inter-judge agreement. To separate the *instrument* effect from a *rater-population* effect (physicians vs LLMs), we additionally ran the same blinded panel under the **pairwise** instrument, filling three of the four cells of a {pairwise, rubric} × {human, LLM} design and decomposing the swing into rater-modality (B–A) and instrument-format (C–B) components.

Results. Swapping only the instrument **eliminated OE's advantage on every axis, though not uniformly**: it reversed OE's sign on two axes, collapsed it to statistical null on two, and shrank it roughly threefold — while leaving it significantly positive — on the fifth. On accuracy — the marquee axis — the LLM-rubric panel scored OE **−29.1 pp (95% CI −38.0 to −19.8): a *significantly OE-negative* verdict on the very answers human raters preferred OE on** (+24.4 pp on the full Real-POCQi data; +10.4 pp, CI −1.6 to +22.2, on the identical 150-question subsample, which is same-signed but underpowered). The negative sign survived dropping any single judge, including the self-preferring GPT-5.5 (−12.3 pp without it). Clinical utility also reversed significantly; **source quality kept a significantly positive but ~3× smaller OE edge (+12.0 pp, CI excluding zero)**; completeness and verifiability collapsed to statistical null. GPT-5.5 self-preferred (+0.481 points to own family; Opus +0.121; Gemini +0.004), amplifying but not creating the reversal. Inter-judge agreement was modest (Spearman 0.19–0.47), mirroring the low rater agreement reported by the Nature study itself. **The 2×2 decomposition (cell A computed on the *same* questions as B and C) localizes the cause to the instrument, not the rater: LLM judges administering the *pairwise* instrument reproduce the human verdict — OE wins on four of five axes — and the instrument-format component (C–B, −30 to −44 pp on every axis) dominates the rater-modality component (B–A). On accuracy specifically, the instrument component (−32.7 pp) accounts for ~83% of the human-pairwise→LLM-rubric swing; rater**

modality contributes only −6.8 pp. The reversal is also **not a length artifact**: OE produces the *longest* answers yet loses on accuracy, and length-adjusting the rubric gap leaves it unchanged within CI.

Conclusions. The two studies' disagreement is driven substantially by the **evaluation instrument** (pairwise preference vs absolute rubric) rather than by whether humans or LLMs do the rating, independently of query provenance, model version, and answer content. This conclusion is established *within LLM raters* (cells B and C); bridging it to the human-rated Nature rubric requires assuming no rater×instrument interaction — i.e. that human raters would show the same pairwise→rubric shift LLMs do. We could **not** fill the fourth {human, rubric} cell, so we flag this additivity assumption as the central untested step and the primary target for follow-up (even a small human-rubric sample would test it). Subject to that caveat, "which clinical AI is best" is not instrument-invariant; leaderboards must report the instrument as a first-class experimental factor. We additionally document two under-reported flaws common to both source studies — omission of the actual product clinicians use (the ChatGPT product), and unreported/uncontrolled reasoning effort — and pre-register a full provenance × instrument × citation factorial (including the missing human-rubric cell) to estimate each factor's causal contribution.

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1. Introduction

Point-of-care clinical AI is now evaluated by head-to-head studies whose results directly influence purchasing and deployment. Medical-AI evaluation has grown rapidly, spanning licensing-exam question banks,[1] physician-aligned rubric benchmarks,[2] large language models that encode clinical knowledge,[3] specialized and mobile-sized clinical models,[4,5,6] health-system-scale prediction and operations models,[7,8] sequential diagnostic reasoning,[9] retrieval-augmented generation for healthcare,[10] and a parallel literature documenting these systems' failure modes — susceptibility to distraction,[11] conflict between an LLM's internal prior and retrieved evidence,[12] data-poisoning attacks,[13] and the need for clinically safe generation.[14] These benchmarks increasingly characterize their query distributions with topic models.[15] Deployment is already at health-system scale.[16] In 2026 two such studies reached **opposite** conclusions using overlapping systems,[17,18] which is the strongest possible signal that at least one conclusion is an artifact of method rather than of the systems compared.

Both studies contain a symmetric structural feature — each sourced its "real-world" query set from the platform that ultimately won (Real-POCQi from OpenEvidence traffic; the Nature study's real-clinical- query benchmark from an NYU Langone GPT deployment) — and each used a **different evaluation instrument** (blinded human pairwise preference vs absolute rubric scoring). Provenance and instrument are therefore fully confounded in the published record. This paper isolates the instrument by holding everything else constant, and then situates that result within a fuller critique and a pre-registered factorial design.

Our contributions: 1. **An instrument existence proof plus a rater/instrument decomposition** (executed, on public data): with queries and answers held fixed, changing the instrument reverses, nullifies, or sharply attenuates OE's advantage on every axis (§4.1); and by filling three cells of a {pairwise, rubric} × {human, LLM} design we show the swing is driven by the **instrument format**, not by the human-vs-LLM rater change — LLMs given the pairwise instrument agree with physicians on four of five axes (§4.5). 2. **Two methodological critiques** of both source studies: the comparator set omits the ChatGPT product clinicians actually use, and reasoning effort is unreported or uncontrolled (§5). 3. **A pre-registered 2×2×2** provenance × instrument × citation design (§7) that would estimate each factor's causal contribution, with power grounded in measured variance components. This is a *proposed design, not an executed result* — we flag it as such to keep the executed contribution (#1) cleanly separated from future work.

2. The two source studies (verified)

Numbers below were verified for this work from primary sources (Nature PDF Methods; Real-POCQi abstract + HTML full text).

Real-POCQi[17] (arXiv:2606.28960; dataset `jjfenglab/Real-POCQi`, CC BY 4.0). 620 real point-of-care queries sourced from OpenEvidence[19] plus 187 HealthBench[2] items; 149 physicians across 36 states; blinded **pairwise** comparisons — the arena-style preference paradigm[20] — on five axes (accuracy, clinical utility, source quality, completeness, verifiability). Systems: Claude Opus 4.8, Gemini 3.1 Pro, GPT-5.5, OpenEvidence, all queried via API; temperature 0.0, seed 42, web search enabled; **"Thinking was automatically determined by the LLM."** OE won by roughly +25 to +39 pp win-difference. *The data collection was run by the platform under study (conflict of interest).*

Nature Medicine[18] (s41591-026-04431-5, NYU Langone + UT Austin). MedQA[1] (500) + HealthBench[2] (500) + a 100-item real-clinical-question (RCQ) benchmark sampled from NYU's HIPAA-compliant GPT instance;[16] 12 clinician raters; **absolute 1–4 rubric**. Systems: GPT-5.2 (2025-12-11), Gemini 3.1 Pro Preview, Claude Opus 4.6 (all API), OpenEvidence and UpToDate Expert AI[21] (browser), plus Google Search AI Overview (RCQ). Temperature 0.0, seed 62, search enabled; **reasoning effort not reported.** Length not normalized (by explicit choice). Frontier LLMs won: e.g. HealthBench GPT 88.0 / Gemini 79.3 / Claude 77.0 / OE 62.6 / UpToDate 61.3; RCQ Gemini 3.62 / GPT 3.54 / Claude 3.52 / OE 3.24. HealthBench was graded by an LLM-judge panel (self-preference risk); RCQ item-level agreement was low (Krippendorff $\alpha \approx$ 0.10–0.20). License CC BY-NC-ND 4.0.

The two studies thus differ simultaneously in **provenance, instrument, model version, and access path** — none of which is individually identified by comparing their published leaderboards.

3. Methods

3.1 Design: hold queries and answers fixed, vary only the instrument

We reuse Real-POCQi's public artifacts unchanged: the 620 queries, the four systems' verbatim answers, and the blinded human pairwise ratings. On a seed-fixed sample of **150 queries** (`random_state=62`; 600 answers) we administer a **second instrument** — Nature-style absolute 1–4 rubric scoring — to the *same* answers, then derive the *same* OE-vs-rest win-difference metric from both instruments. Because queries, answers, provenance, and model versions are identical across the two instruments, any change in the metric is attributable to the instrument.

3.2 The rubric instrument (LLM-judge panel)

Each answer is scored 1–4 (1 unacceptable … 4 excellent) on the five Real-POCQi axes by a blinded panel of four judges — GPT-5.5, Claude Opus 4.8, Grok-4.3, Gemini-3.5-flash — chosen to span vendors and to include contestant families (enabling a self-preference measurement) and one family with no contestant (Grok). We stress that Grok is a *family-neutral* reference, **not a bias-free** one: as an LLM it may still share the panel's general preference for frontier-style prose (structure, breadth, hedging), a house effect that leave-one-judge-out cannot remove (§4.2, Limitation iii). Judges see only the question and one answer, are told nothing about the source system, and return JSON scores only (system prompt in `judge/grade.py`). All judges run at **high reasoning effort**; we **verify** reasoning-token consumption on a real grading item (GPT-5.5 3,071; Grok-4.3 1,880; Gemini-3.5-flash 922; Opus-4.8 443 thinking tokens; `judge/verify_thinking.py`). 2,388 of 2,401 answer-grades completed (0.5% failures — transient timeouts and a few Gemini truncations — balanced across systems).

3.3 Metric and statistics

For each axis we compute the **panel-mean 1–4 score per (question, system)**, then form the OE-vs-rest **win-difference** = 100 × (wins − losses) / comparisons across the three frontier systems per question — the same one-vs-rest statistic Real-POCQi reports for human pairwise, a win-ratio-family contrast[22] in the arena-style preference tradition.[20] Uncertainty: 2,000 cluster-bootstrap replicates resampling **questions** (cluster on `question_id`). Robustness: **leave-one-judge-out** (recompute dropping each judge). Bias: **self-preference** = mean own-family minus mean others, paired within (question, axis). **Inter-judge agreement** = Spearman correlation of per-(question, system) mean scores. Human comparison uses the `qa_text_only` render mode (matching the citation-free OE answers in the dataset).

Multiplicity. We test five axes; we report per-axis 95% cluster-bootstrap CIs without a formal family-wise correction, so the two axes closest to the null boundary (source quality lower CI +1.6; clinical utility upper CI −3.6) would not necessarily survive Bonferroni/Holm adjustment across five tests. The headline accuracy reversal (CI −38.0 to −19.8) is far from the boundary and robust to any standard correction. We flag the borderline axes explicitly rather than over-claiming per-axis significance.

3.4 The pairwise cell (2×2 decomposition)

To separate the instrument from the rater population we administer a **second instrument to the same LLM panel**: blinded forced-choice **pairwise** preference (A/B/tie per axis), OpenEvidence vs each of the three frontier systems, on the same 150-query sample (`judge/pairwise.py`). Slot order is deterministically randomized per (question, opponent, judge) via a hash and de-blinded at scoring, removing position bias. We de-blind each A/B/tie verdict to an OE win/loss/tie and compute the identical OE-vs-rest win-difference, with the same 2,000-replicate `question_id` cluster bootstrap. This yields cell **B = {pairwise, LLM}**, which — with cell A = {pairwise, human} (Real-POCQi) and cell C = {rubric, LLM} (§3.2) — gives three of the four cells of the {pairwise, rubric} × {human, LLM} design and identifies the **rater-modality effect (B−A)** and the **instrument-format effect (C−B)** (§4.5). Judges run at high reasoning effort; all four judges completed 448–450/450 comparisons (1,798/1,800 overall).

4. Results

4.1 The instrument flips the winner (Figure 1, Table 1)

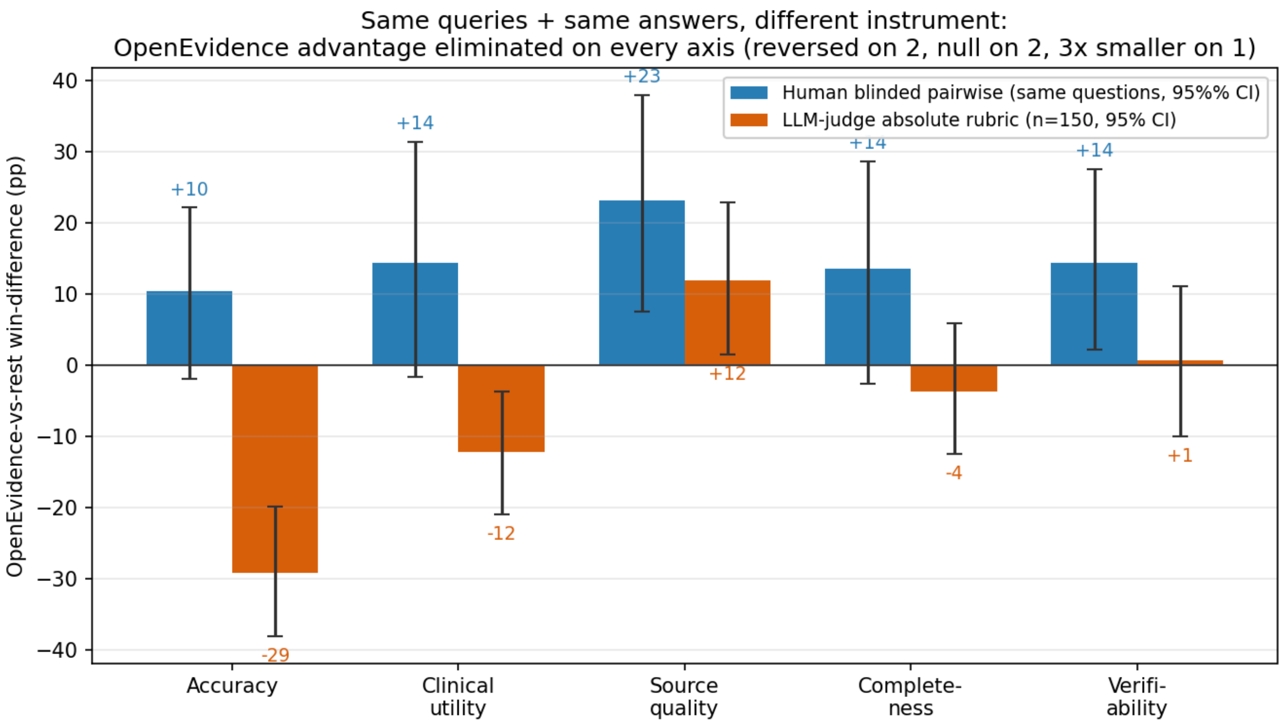

Figure 1. Instrument existence proof. OE-vs-rest win-difference (percentage points) per axis under the human blinded pairwise instrument (same 150 questions) versus the LLM-judge absolute-rubric panel (n = 150), with 95% cluster-bootstrap confidence intervals. Holding queries and answers fixed and changing only

the instrument eliminates OE's advantage on every axis — reversed on two, null on two, threefold smaller on one.

Table 1. OE-vs-rest win-difference (pp) by axis under each instrument, **both computed on the same 150-query sample** (2,000-replicate `question_id` cluster bootstrap). The human column is the human pairwise win-difference restricted to the identical questions (86–119 of 150 carry human text-only ratings, by axis); the "full data" column is the win-difference on Real-POCQi's complete text-only bank (reproduced in §4.4), shown for reference because the subsample is underpowered. The native-gap column gives the OE-minus-mean-of-frontier gap on the raw 1–4 rubric (see scale note).

Axis	Human pairwise, same 150 (pp) [95% CI]	Human, full data (ref)	LLM-judge panel (pp) [95% CI]	Native gap (1–4 pts)	Verdict
Accuracy	+10.4 [−1.8, +22.2]	+24.4	**−29.1 [−38.0, −19.8]**	−0.127	LLM rubric significantly negative; sign flips
Clinical utility	+14.4 [−1.6, +31.4]	+29.5	−12.2 [−20.9, −3.6]	−0.021	LLM rubric significantly negative; sign flips
Source quality	+23.2 [+7.6, +38.0]	+38.1	+12.0 [+1.6, +22.9]	+0.103	OE edge survives, ~3× smaller
Completeness	+13.6 [−2.5, +28.7]	+30.3	−3.6 [−12.4, +6.0]	+0.004	collapses to null
Verifiability	+14.4 [+2.2, +27.6]	+25.5	+0.7 [−10.0, +11.1]	+0.003	collapses to null

The headline is accuracy. Real-POCQi's central finding is that physicians prefer OE on accuracy (+24.4 pp on the full data; +10.4 pp, CI −1.8 to +22.2, on this 150-query subsample — same sign, but the subsample carries only 86 accuracy ratings and is underpowered). On the **same answers**, the LLM-rubric panel scores OE **−29.1 pp [−38.0, −19.8] — significantly negative**. So the instrument swap moves the accuracy verdict from OE-favoring (significantly so at full power; positive but CI-crossing in this subsample) to *significantly OE-disfavoring*. Two axes flip sign into significantly-negative territory (accuracy, clinical utility), one retains a much-attenuated but significantly positive OE edge (source quality), and two collapse to null. Under no axis does OE retain its large human-pairwise advantage. We deliberately avoid framing this as "a significant +24 becomes a significant −29": at the 150-query existence-proof scale the human accuracy estimate is not itself significant, so the clean claim is a **sign reversal to a significantly-negative rubric verdict on identical content**, corroborated by the full-data human sign.

Note on scale (native vs win-difference). The win-difference is a sign-of-the-gap statistic: because per-question frontier scores cluster tightly, a mean rubric gap as small as **−0.127 of one point** (accuracy) is enough to flip the majority of head-to-head comparisons and produce a −29 pp win-difference, while a near-zero gap (completeness +0.004, verifiability +0.003) yields a null win-difference. We therefore report both scales: the **native gap** shows the *effect size is modest in absolute terms*, and the **win-difference** shows it is *directionally decisive* under the same one-vs-rest rule Real-POCQi used. Neither instrument's metric should be read as a large clinical-quality gulf; the point is that the two instruments order the systems differently on identical content.

4.2 The reversal is not merely a self-scoring artifact

LLM-as-judge panels are now a standard evaluation tool[23] but are known to favor their own generations,[24] so we measure this directly. Contestant-family judges self-preferred (GPT-5.5 **+0.481**, Opus **+0.121**, Gemini **+0.004** points, own family minus others). Yet leave-one-judge-out shows the **accuracy reversal survives dropping any single judge, including GPT-5.5** (−12.3 pp with GPT-5.5 removed — still negative). GPT-5.5 amplifies the flip roughly fourfold but does not create it. By contrast, the clinical-utility reversal is GPT-5.5-dependent (→ +2.5 pp without it) and is reported as such.

A residual house effect remains, however. Leave-one-judge-out and the self-preference statistic only address *family-specific* bias — a judge favoring its own vendor's answer. They do **not** remove the bias all four LLM judges may *share*: a common preference for the structured, comprehensive, frontier-style prose that general-purpose LLMs produce (and are trained on), against OE's terser clinician-tuned framing. Because even our family-neutral judge (Grok) is itself an LLM, no judge in the panel is a clean control for this shared stylistic prior. The frontier "win" under the rubric is therefore best read as *"frontier answers score higher on an LLM-administered rubric,"* not as a family-free verdict of clinical superiority — a caution we carry into the Discussion and Limitation (iii). The LLM-pairwise cell (§4.5) directly tests this house effect and finds it is **specific to the rubric instrument**: the same judges — including the family-neutral one — prefer OE under pairwise. Human- administered rubric scoring would be the remaining control (pre-registered, §7).

4.3 The instrument is noisy

Inter-judge Spearman agreement was modest (0.19–0.47), closely mirroring the low item-level agreement the Nature study reports for its own human raters ($\alpha \approx 0.10$–0.20). Rubric scoring — whether by humans or LLMs — is a higher-variance instrument than forced-choice preference; this is part of *why* it can diverge from pairwise, not a defect unique to our panel.

4.4 Public-data reproductions (pipeline validation)

Our one-vs-rest win-difference on the human text-only data reproduces Real-POCQi to <1 pp on every axis (e.g., accuracy +24.4 vs 24.7 published). A separately reported citation-halo analysis (`analysis/analyze.py`) finds OE's between-groups accuracy "halo" (+11.4 pp) **collapses to +1.5 pp (NS)** within questions rated in both render modes — i.e., largely a selection artifact, not a causal citation effect — motivating the randomized citation arm in §7. Measured question-level SD = 0.598 grounds the power analysis.

4.5 Decomposing the swing: it is the instrument, not the rater (Figure 2, Table 2)

A natural objection to §4.1 is that swapping {human pairwise} → {LLM rubric} changes **two** things at once — the *rater population* (physicians → LLMs) and the *instrument format* (forced-choice preference → absolute rubric) — so the reversal could be a human-vs-LLM effect rather than an instrument effect. The two source studies share exactly this confound. We break it by filling the missing cell of the 2×2 {pairwise, rubric} × {human, LLM} design: we ran the **same four-judge LLM panel, blinded, at high reasoning effort, on the same 150 queries and answers**, but administered the **pairwise** instrument (forced A/B/tie per axis, slot order deterministically randomized and de-blinded at scoring; `judge/pairwise.py`). This gives three of four cells — A = {pairwise, human} (Real-POCQi), B = {pairwise, LLM} (new), C = {rubric, LLM} (§4.1) — and lets us attribute the A→C swing to a **rater-modality component (B–A)** and an **instrument-format component (C–B)**.

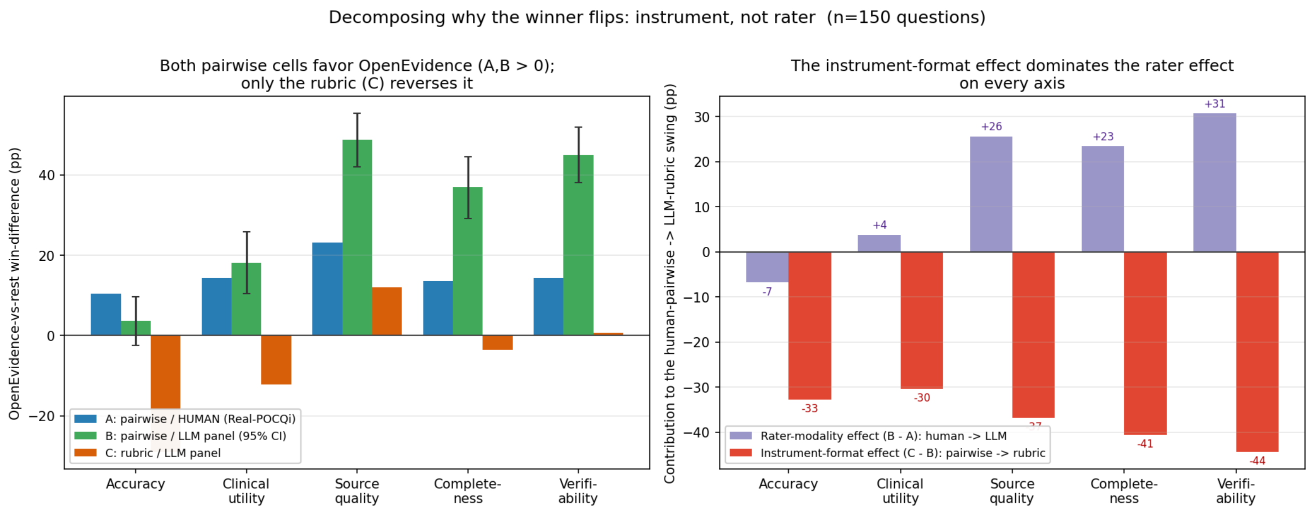

Figure 2. The 2×2 rater-vs-instrument decomposition (n = 150). *Left:* three cells of the {pairwise, rubric} × {human, LLM} design per axis — both pairwise cells (A: human; B: LLM) favor OpenEvidence; only the rubric cell (C) reverses it. *Right:* the human-pairwise→LLM-rubric swing split into a rater-modality component (B−A) and an instrument-format component (C−B); the instrument component dominates on every axis.

Table 2. OE-vs-rest win-difference (pp) across three cells of the 2×2, and the decomposition of the human-pairwise→LLM-rubric swing. **Cell A is computed on the same 150-query sample as B and C** (human text-only ratings restricted to those questions), *not* on the full text-only bank — otherwise B−A would absorb a sample-composition difference into the "rater" term. Cell B: n=150, 2,000-replicate cluster bootstrap; 8,990 axis-verdicts (1,798/1,800 comparisons, all four judges 448–450/450).

Axis	A: pairwise/human (same 150)	B: pairwise/LLM [95% CI]	C: rubric/LLM	Total (C−A)	Rater (B−A)	Instrument (C−B)
Accuracy	+10.4	+3.6 [−2.5, +9.6]	−29.1	−39.5	−6.8	**−32.7**
Clinical utility	+14.4	+18.2 [+10.4, +25.9]	−12.2	−26.6	+3.8	**−30.4**
Source quality	+23.2	+48.8 [+42.1, +55.4]	+12.0	−11.2	+25.6	**−36.8**
Completeness	+13.6	+37.0 [+29.2, +44.5]	−3.6	−17.2	+23.4	**−40.6**
Verifiability	+14.4	+45.1 [+38.1, +51.9]	+0.7	−13.7	+30.7	**−44.4**

Two results stand out. **First, LLM judges administering the *pairwise* instrument largely reproduce the human pairwise verdict**: OE wins on four of five axes (source quality +48.8, verifiability +45.1, completeness +37.0, clinical utility +18.2), with only accuracy attenuating to a null +3.6. The much-feared "LLMs simply disagree with physicians" story is therefore false for four of five axes — given the *same* forced-choice instrument, LLMs and physicians agree that OE's answers are preferred. **Second, the instrument-format component (C−B) is large, negative, and consistent on every axis (−30.4 to −44.4 pp), and it dominates the rater-modality component (B−A, which is small on accuracy and positive on three axes).** The swing from "OE wins" to "OE loses" is thus attributable primarily to the **pairwise→rubric instrument change**, not to the human→LLM rater change. Accuracy is the one axis where both components push in the same (negative) direction, but even there the instrument does the overwhelming majority of the work: **−32.7 pp instrument vs only −6.8 pp rater — ~83% of the swing is the instrument.** (With cell A taken instead on the full text-only bank, the accuracy rater term would appear as −20.8 pp; that larger value is an artifact of comparing the LLM cells against a different, larger question sample, and we do not use it.)

This also sharpens the house-effect discussion (§4.2). Under the pairwise instrument, the **family-neutral judge (Grok) prefers OE on all five axes** (+11 to +65 win-diff) and the contestant **GPT-5.5 is the *least* OE-favorable** judge — the *opposite* of what a shared pro-frontier stylistic bias would predict. The pro-frontier tilt is therefore specific to the **rubric** instrument, not a blanket LLM prejudice against OE. (One robustness note: Gemini-3.5-flash initially stalled at 297/450 pairwise verdicts when the Google API account exhausted its credit quota mid-run; after credits were restored we completed the cell to 448/450, and the point estimates shifted <3 pp from the partial-coverage run — i.e., the conclusion never depended on the missing data.)

4.6 The rubric reversal is not a length artifact (Table 3)

Answer length is the most-cited uncontrolled confound in both source studies (neither normalizes length). The natural worry is that the rubric rewards longer, more comprehensive answers. We test this on the rubric cell without regenerating any answers (`judge/length_analysis.py`). The premise fails at the first step: **OpenEvidence produces the *longest* answers** (median 3,600 chars vs GPT-5.5 2,232, Opus 3,294, Gemini 3,586), yet it *loses* the accuracy rubric — so a "longer answers win" mechanism runs *backwards* against the finding. We confirm this three ways (Table 3): (i) the per-axis length **slope** is tiny (−0.015 to +0.042 rubric points per 1,000 characters); (ii) the **length-adjusted intercept** — the expected OE-minus-frontier score gap at *equal length* — is statistically indistinguishable from the raw gap on every axis (accuracy −0.12 [−0.17, −0.08] adjusted vs −0.127 raw); and (iii) the accuracy win-difference stays negative **whether OE's answer is longer (−34.3) or shorter (−19.9) than its opponent's** — OE loses on accuracy even when it is the longer answer.

Table 3. Length sensitivity of the rubric cell (n=150; 2,000-replicate cluster bootstrap). Native-scale OE-minus-frontier score gap: raw vs length-adjusted (intercept at equal length); and the accuracy win-difference split by whether OE is the longer or shorter answer.

Axis	Raw gap (pts)	Length-adjusted gap [95% CI]	Slope (pts / +1k chars)	Win-diff: OE longer	Win-diff: OE shorter
Accuracy	−0.127	−0.12 [−0.17, −0.08]	−0.015	−34.3	−19.9
Clinical utility	−0.021	−0.03 [−0.06, +0.01]	+0.016	−14.1	−8.4
Source quality	+0.103	+0.09 [+0.03, +0.15]	+0.042	+15.9	+6.0
Completeness	+0.004	−0.01 [−0.05, +0.03]	+0.039	+0.7	−10.8
Verifiability	+0.003	−0.01 [−0.06, +0.05]	+0.025	+1.4	+0.0

Length has a small positive association with source-quality/completeness/verifiability scores (slopes +0.03 to +0.04 per 1,000 chars) but essentially none with accuracy, and adjusting for it leaves every axis's OE-vs-frontier gap unchanged within CI. The accuracy reversal is therefore **not** explained by length. (This does not make length irrelevant in general — a fully length-matched *generation* study is still pre-registered, §7 — but it removes length as an alternative explanation for the observed reversal.)

5. Critiques of both source studies

The two critiques below are **independent of the instrument existence proof** (§4): they hold whether or not the instrument drives the disagreement, and they apply symmetrically to both source studies. We include them because they bound how far *either* published leaderboard can be trusted for procurement, but a reader interested only in the instrument result can treat this section as standalone context. Full detail in `CRITIQUES.md`. Two points bear directly on interpretation:

(a) The comparator set omits the AI clinicians actually use. Both studies benchmark **bare frontier API models**; neither includes the **ChatGPT product** (consumer or clinician deployment), whose system prompt, retrieval/browsing, memory, and safety guardrails move the very axes under test. This is especially asymmetric because OE is itself a curated clinical *product*; Nature further queries clinical tools via browser but frontier models via API. A fair benchmark should include a **product arm** distinct from the raw model endpoint.

(b) Reasoning effort is unreported or uncontrolled. Nature reports temperature 0.0 and a seed but **never states reasoning effort** (its cost table confirms reasoning tokens were spent, at an unspecified level); Real-POCQi reports only that "thinking was automatically determined by the LLM." Reasoning effort is the largest controllable performance lever for these models and can reorder leaderboards, so neither head-to-head is reproducible on this axis. Our study pins high effort and verifies token consumption (§3.2).

A corroborating confound: Real-POCQi tested **newer** frontier weights (GPT-5.5, Opus 4.8) than Nature (GPT-5.2, Opus 4.6) and still found them losing on human pairwise — a pattern more consistent with an instrument effect than a model-version effect.

6. Discussion

Holding queries and answers fixed, the instrument alone is sufficient to reverse the central claim of a published clinical-AI benchmark. The 2×2 decomposition (§4.5) makes this attribution explicit rather than inferential: because the *same* LLM panel that reverses OE under the rubric **reproduces the human pairwise verdict when it uses the pairwise instrument** (OE winning on four of five axes), the swing localizes to the pairwise→rubric change (C–B, −30.4 to −44.4 pp) and not to the human→LLM change (B–A).

Two scoping caveats bound this claim. *First*, the decomposition is established **within LLM raters**: we have cells A = {pairwise, human}, B = {pairwise, LLM}, and C = {rubric, LLM}, but **not** the fourth cell D = {rubric, human}. The two-study reconciliation is ultimately a claim about a *human-rated* rubric study (Nature), so generalizing our within-LLM instrument effect (C–B) to explain Nature requires assuming **no rater×instrument interaction** — that physicians administering a rubric would show the same pairwise→rubric shift LLMs do. That assumption is plausible but untested, and this is precisely the direction where interaction is likely: humans may not share the frontier-prose prior that an LLM-scored rubric appears to reward, so the human-rubric cell could differ materially from our LLM-rubric cell. We therefore state the proven claim narrowly ("the instrument dominates the rater *modality* among LLM raters") and treat the bridge to Nature as conditional on additivity; filling cell D — even with a small human-rubric sample — is the single highest-value follow-up. *Second*, the axis where our headline lives, accuracy, is also the axis where the rater term is not negligible in the same direction: instrument −32.7 pp and rater −6.8 pp (≈83% instrument), cleaner than before once cell A is put on the same sample, but not a pure instrument effect. The clean "instrument, not rater" statement is strongest for source quality/verifiability/completeness and merely dominant — not exclusive — for accuracy.

This does not show OE is worse (or better) than frontier models in the clinic — we have **no adjudicated ground truth** — but it shows that "which system is best" is a function of the measuring instrument, and that pairwise-preference and absolute-rubric instruments encode materially different value functions (preference rewards OE's clinician-tuned framing and citations; rubric scoring rewards frontier models' breadth and structure). It also reframes the "LLM-judge self-preference" worry: the pro-frontier tilt is a property of the **rubric instrument**, not of LLM judges per se, since those same judges favor OE under pairwise — including the family-neutral judge. Benchmarks that do not report the instrument as an experimental factor — and that do not control reasoning effort, comparator product-vs-endpoint status, and length — are under-specified for procurement decisions.

7. Pre-registered extension

The executed existence proof identifies the instrument; a full **2×2×2 provenance × instrument × citations** factorial (protocol in `reconciliation_protocol.md`) estimates each factor's causal share, including a randomized citation-halo arm (motivated by §4.4), a length-matched *generation* sub-study (to confirm causally what §4.6 shows observationally — that length does not drive the reversal), a product-vs-endpoint arm (§5a), and reasoning-effort sweeps (§5b). Power is grounded in the measured question-level SD (0.598): ≈250 questions/cell for a 15 pp interaction, ≈390/cell for 12 pp. Because the Nature RCQ corpus is not public, the provenance arm uses a constructed surrogate LLM-platform query corpus; Real-POCQi is directly reusable.

8. Limitations

(i) **No ground truth** — we show instruments disagree, not which is correct. (ii) **Length not normalized** (OE 3.8k vs GPT 2.6k chars) — we show *observationally* (§4.6) that the reversal is not a length artifact (OE is the longest provider yet loses; length-adjusted gaps match raw gaps within CI), but a fully length-matched *generation* study is still pre-registered to settle it causally. (iii) **House effects** — contestant-family judges self-prefer (quantified; controlled via leave-one-judge-out), and the panel may share a rubric-specific pro-frontier stylistic prior that leave-one-judge-out cannot remove; the pairwise cell (§4.5) shows this prior is instrument-specific (the same judges favor OE under pairwise), and the remaining control — human-administered rubric scoring — is pre-registered. (iv) **Noisy instrument** — modest inter-judge agreement. (v) **Missing fourth cell and the additivity assumption** — we have three of the four 2×2 cells; the decomposition is therefore identified *within LLM raters*, and bridging the instrument effect to the human-rated Nature rubric assumes **no rater×instrument interaction** (that physicians doing a rubric would show the same pairwise→rubric shift LLMs do). This is untested and is the likely direction of interaction (humans may not share the frontier-prose prior); filling cell D = {rubric, human}, even at small n, is the top follow-up. Relatedly, cell A is Real-POCQi's human pairwise protocol while cell B is our LLM pairwise reimplementation, so the rater term B−A absorbs minor **protocol** differences (tie handling, pair structure) as well as rater modality — it is "rater modality + protocol," not pure rater. Both cell A and cell B are computed on the identical 150-query sample, so B−A is at least free of sample-composition confounding. The n=150 existence-proof scale is decisive for the LLM-rubric verdict (cell C significantly negative on accuracy and clinical utility) but underpowers the same-sample human baseline (three axes' human CIs cross zero in the subsample, though the full-data values are significant and same-signed); the null-collapsing axes await the full run. (vi) **No rater-level modeling** — the public ratings table has no rater identifier, so human-side rater random effects cannot be fit from public data. (vii) **LLM judges are not bit-deterministic**; we quantify rather than assume stability.

9. Ethics, conflicts, funding

No human subjects or PHI were used; all human ratings are de-identified public data (CC BY 4.0). No Nature-copyrighted data were redistributed (CC BY-NC-ND). **Conflicts of interest:** The author is the founder of Kinvectum AB, an independent medical-AI research venture. The author declares no financial interest in, and no commercial or consulting relationship with, any of the evaluated systems or their providers (OpenEvidence, OpenAI, Anthropic, xAI, Google, or Wolters Kluwer/UpToDate). For transparency we note the source studies' own disclosed conflicts: Real-POCQi's data collection was run by OpenEvidence (the platform under study), and the Nature Medicine senior author reports industry equity/consulting including Google. **Funding:** This work was funded by Kinvectum AB, which covered the LLM-judge API costs; the funder had no role in study design, analysis, or the decision to publish. **Author contributions:** K.A. designed the study, wrote the analysis and judging code, performed all analyses, and wrote the manuscript (sole author).

10. Data and code availability

All code and outputs are in this repository; `run_all.sh` reproduces every result end to end (`SKIP_JUDGES=1` for the no-API-key public-data subset). Real-POCQi data: Hugging Face `jjfenglab/Real-POCQi` (CC BY 4.0), fetched by `fetch_data.py` and vendored in `data/`. Judge configuration, reasoning-token verification, grades, bootstrap CIs, and figures are regenerable as documented in `README.md`. Nature Medicine numbers are cited from the published article (s41591-026-04431-5); its RCQ corpus is not public (IRB i23-00510).

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Citations appear as numbered footnotes at the point of use and are collected in full below. Numbering follows order of first appearance (Vancouver style). Full author lists and target-journal formatting to be finalized before submission.

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